

8. SQLite Database - Create Employee table with at least 4 fields of different data types (eno, name, gender, salary), Perform the following operations: Create, Insert, show all records(in the next activity), Search (based on eno), delete (based on name of the employee), and Update (based on name of the employee).

MainActivity.java:

```
package com.example.myapplication;
import android.app.NotificationChannel;
import android.app.NotificationManager;
import android.app.PendingIntent;
import android.content.Context;
import android.content.Intent;
import android.database.Cursor;
import android.net.Uri;
import android.os.Build;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.app.NotificationCompat;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
import java.io.BufferedReader;
import java.io.File;
import java.io.FileInputStream;
import java.io.FileOutputStream;
import java.io.InputStreamReader;
public class MainActivity extends AppCompatActivity {
    EditText edteno, edtname, edtgender, edtsalary;
    Button btnadd, btnshow, btnsearch, btndel, btnupd;
    DBHelper conn;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
            return insets;
        });
        edteno = findViewById(R.id.edteno);
        edtname = findViewById(R.id.edtname);
        edtgender = findViewById(R.id.edtgender);
        edtsalary = findViewById(R.id.edtsalary);
        btnadd = findViewById(R.id.btnadd);
        btnshow = findViewById(R.id.btnshow);
        btnsearch = findViewById(R.id.btnsearch);
```

```
btndel = findViewById(R.id.btndel);
btnupd = findViewById(R.id.btnupd);
conn = new DbHelper(this);
conn.getWritableDatabase();
// Insert
btnadd.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        int e = Integer.parseInt(edteno.getText().toString());
        String n = edtname.getText().toString();
        String g = edtgender.getText().toString();
        float s = Float.parseFloat(edtsalary.getText().toString());
        boolean b = conn.insertData(e, n, g, s);
        if (b) {
            Toast.makeText(MainActivity.this,
                "Record Inserted Successfully",
                Toast.LENGTH_LONG).show();
        } else {
            Toast.makeText(MainActivity.this,
                "Insert Failed",
                Toast.LENGTH_LONG).show();
        }
    }
});
// Show all records in next activity
btnshow.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        Intent i = new Intent(MainActivity.this, ResultActivity.class);
        startActivity(i);
    }
});
// Search by eno
btnsearch.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        int e = Integer.parseInt(edteno.getText().toString());
        Cursor c = conn.searchData(e);
        if (c.getCount() == 0) {
            Toast.makeText(MainActivity.this,
                "No Record Found",
                Toast.LENGTH_LONG).show();
        } else {
            if (c.moveToFirst()) {
                edteno.setText(c.getString(0));
                edtname.setText(c.getString(1));
                edtgender.setText(c.getString(2));
                edtsalary.setText(c.getString(3));
            }
        }
        c.close();
    }
});
```

```
    }  
});  
// Delete by name  
btndel.setOnClickListener(new View.OnClickListener() {  
    @Override  
    public void onClick(View v) {  
        String n = edtname.getText().toString();  
        int rows = conn.deleteData(n);  
        if (rows != 0) {  
            Toast.makeText(MainActivity.this,  
                "Deleted Successfully",  
                Toast.LENGTH_LONG).show();  
        } else {  
            Toast.makeText(MainActivity.this,  
                "Delete Failed",  
                Toast.LENGTH_LONG).show();  
        }  
    }  
});  
// Update by name  
btnupd.setOnClickListener(new View.OnClickListener() {  
    @Override  
    public void onClick(View v) {  
        String n = edtname.getText().toString();  
        int e = Integer.parseInt(edteno.getText().toString());  
        String g = edtgender.getText().toString();  
        float s = Float.parseFloat(edtsalary.getText().toString());  
        int rows = conn.updateData(n,e, g, s);  
        if (rows != 0) {  
            Toast.makeText(MainActivity.this,  
                "Updated Successfully",  
                Toast.LENGTH_LONG).show();  
        } else {  
            Toast.makeText(MainActivity.this,  
                "Update Failed",  
                Toast.LENGTH_LONG).show();  
        }  
    }  
});  
}
```

ResultActivity.java:

```
package com.example.myapplication;  
  
import android.database.Cursor;  
  
import android.os.Bundle;
```

```
import android.widget.TextView;

import android.widget.Toast;

import androidx.activity.EdgeToEdge;

import androidx.appcompat.app.AppCompatActivity;

import androidx.core.graphics.Insets;

import androidx.core.view.ViewCompat;

import androidx.core.view.WindowInsetsCompat;

public class ResultActivity extends AppCompatActivity {

    TextView tv;

    DbHelper conn;

    @Override

    protected void onCreate(Bundle savedInstanceState) {

        super.onCreate(savedInstanceState);

        EdgeToEdge.enable(this);

        setContentView(R.layout.activity_result);

        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {

            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());

            v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);

            return insets;

        });

        tv = findViewById(R.id.tv);

        conn = new DbHelper(this);

        Cursor c = conn.showData();

        StringBuffer buf = new StringBuffer();

        if (c.getCount() == 0) {
```

```
Toast.makeText(this,

    "No Records Found",

    Toast.LENGTH_LONG).show();

} else {

    while (c.moveToNext()) {

        buf.append("Employee No.: ")

            .append(c.getString(0))

            .append("\n");

        buf.append("Name: ")

            .append(c.getString(1))

            .append("\n");

        buf.append("Gender: ")

            .append(c.getString(2))

            .append("\n");

        buf.append("Salary: ")

            .append(c.getString(3))

            .append("\n\n");

    }

    tv.setText(buf.toString());

}

c.close();

}

}
```

DbHelper.java:

```
package com.example.myapplication;

import android.content.ContentValues;
import android.content.Context;
import android.database.Cursor;
import android.database.sqlite.SQLiteDatabase;
import android.database.sqlite.SQLiteOpenHelper;

public class DbHelper extends SQLiteOpenHelper {

    public static final int DB_VERSION = 1;

    public static final String DB_NAME = "employee.db";

    public static final String TABLE_NAME = "emp";

    public static final String ENO = "eno";

    public static final String NAME = "name";

    public static final String GENDER = "gender";

    public static final String SALARY = "salary";

    public DbHelper(Context context) {

        super(context, DB_NAME, null, DB_VERSION);

    }

    @Override

    public void onCreate(SQLiteDatabase db) {

        String q = "CREATE TABLE " + TABLE_NAME + " (" +

            ENO + " INTEGER PRIMARY KEY, " +

            NAME + " TEXT, " +

            GENDER + " TEXT, " +

            SALARY + " REAL)";
```

```
        db.execSQL(q);

    }

    @Override

    public void onUpgrade(SQLiteDatabase db, int oldVersion, int newVersion) {

    }

    // Insert

    public boolean insertData(int e, String n, String g, float s) {

        SQLiteDatabase db = getWritableDatabase();

        ContentValues cv = new ContentValues();

        cv.put(ENO, e);

        cv.put(NAME, n);

        cv.put(GENDER, g);

        cv.put(SALARY, s);

        long id = db.insert(TABLE_NAME, null, cv);

        return id != -1;

    }

    // Show all

    public Cursor showData() {

        SQLiteDatabase db = getReadableDatabase();

        String q = "SELECT * FROM " + TABLE_NAME;

        return db.rawQuery(q, null);

    }

    // Search by employee number

    public Cursor searchData(int e) {

        SQLiteDatabase db = getReadableDatabase();
```

```
String q = "SELECT * FROM " + TABLE_NAME +  
    " WHERE " + ENO + " = " + e;  
return db.rawQuery(q, null);  
}  
  
// Delete by name  
public int deleteData(String n) {  
    SQLiteDatabase db = getWritableDatabase();  
    return db.delete(TABLE_NAME, NAME + "=?", new String[]{n});  
}  
  
// Update by name  
public int updateData(String n, int e, String g, float s) {  
    SQLiteDatabase db = getWritableDatabase();  
    ContentValues cv = new ContentValues();  
    cv.put(ENO, e);  
    cv.put(GENDER, g);  
    cv.put(SALARY, s);  
    return db.update(  
        TABLE_NAME,  
        cv,  
        NAME + "=?",  
        new String[]{n}  
    );  
}
```

OUTPUT:**1. Main screen**

Enter employee no

Enter name

Enter gender

Enter salary

Insert Update Delete

Search Show All

2. Insert

101

Employee1

Female

25000

Insert Update Delete

Search Show All

Record Inserted Successfully

3. Show All Records

9:20

Employee No.: 101
Name: Employee1
Gender: Female
Salary: 25000

Employee No.: 102
Name: Employee2
Gender: Male
Salary: 30000

Employee No.: 103
Name: Employee3
Gender: Female
Salary: 28000

4. Update

103

Employee3

Male

28000

Insert Update Delete

Search Show All

Updated Successfully

5. Search

InsertUpdateDelete

SearchShow All

InsertUpdateDelete

SearchShow All

6. Delete

InsertUpdateDelete

SearchShow All

Deleted Successfully

Employee No.: 101
Name: Employee1
Gender: Female
Salary: 25000

Employee No.: 103
Name: Employee3
Gender: Male
Salary: 28000